

ANDREW TRAMUTOLO

LEAD SOFTWARE ENGINEER · IT DIRECTOR · HEAD OF MARKETING

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SUMMARY

One-man army — lead software engineer, systems administrator, and head of marketing. Programmer and systems administrator with ten years shipping commercial software on GNU/Linux. **Sole developer of three cross-platform C++/Qt 6 products** — instrument calibration in use by **hundreds of customers**, multi-channel sensor validation, and an internal ERP — each carried from architecture through release to documentation and support. Integrated PID control into **EPICS** with Brookhaven National Laboratory. Administers **100+ machines** as IT Director alongside the development work, and applies agentic AI tooling to cut delivery cost.

WORK HISTORY

• GEO Calibration

Metrology Company | 2016 – PRESENT

SOFTWARE DEVELOPMENT MANAGER / IT DIRECTOR / HEAD OF MARKETING

- Sole developer of **GEO Validator X**, a commercial cross-platform calibration product in use by **hundreds of customers** (C++/Qt 6, QML) — architecture, interface, real-time graphing, and Excel reporting.
- Sole developer of **Sensor Validator**, a commercial cross-platform product for multi-channel environmental validation across hundreds of sensors (C++/Qt 6).
- Sole developer of **GEO System**, the company's internal ERP (C++/Qt 6, PostgreSQL).
- Partnered with **Brookhaven National Laboratory** on PID controller integration into **EPICS**, the Experimental Physics and Industrial Control System used to operate particle accelerators and large-scale scientific instrumentation.
- Worked closely with our customers to provide bug fixes and custom solutions tailored for their laboratory requirements.
- Administered **100+ machines** as IT Director — GNU/Linux servers & stations, virtualization, ZFS storage, and replicated back-ups for disaster recovery.
- Directed all software development and presented the full product suite at **NCSL International 2026**.
- Lead author of all published documentation — internal and customer-facing manuals covering every software product, hardware unit, and the PID controller.
- Head of marketing for the product line. Produced video content, fliers, documentation, and email campaigns. Led all artistic direction for the website as well as paper and digital fliers.

• JMI Technologies

Managed Service Provider | 2012 – 2015

LEAD SYSTEMS ADMINISTRATOR

- Managed IT and security systems for an enterprise aviation security company — server infrastructure, network security, and disaster recovery across production environments.
- Received notarized clearance for **JFK Airport** to plan, manage, and install security systems.
- Administered GNU/Linux routers and firewalls for **120+ clients**: BIND, dhcpcd, nftables, and Snort IDS/IPS with deep packet inspection. Managed pfSense firewalls, OpenWRT routers, and many other custom network solutions for customers large and small.
- Managed the company-wide Active Directory domain — user provisioning, Group Policy, and access control across all office workstations.
- Supported small-business clients — public schools, security firms, restaurants — on open-source infrastructure, with tier 1–3 support for staff and client production systems.

• Sign Institute & Labs

2001 – PRESENT

LLC

- Taught senior citizens to use their computers and phones — safe web browsing, email, and troubleshooting.
- Produced 3D design, graphic design, and marketing work remotely.

SKILLS

Languages C, C++, C#, Python, QML, SQL, Bash/shell scripting, LaTeX

Frameworks Qt 6, Qt Quick/QML, EPICS, WebAssembly, Godot, Unreal Engine 5

Toolchain & Debugging GCC, Clang/LLVM, CMake, Make, GDB, Valgrind, strace/ltrace, perf, pkg-config

GNU/Linux Internals glibc, IPC, udev, D-Bus, Wayland/X11, systemd

Infrastructure Proxmox, QEMU/KVM, LXC, ZFS, BTRFS, LVM, TrueNAS/FreeNAS, FreePBX, Windows Server

Networking & Security Linux routing, TCP/IP, DNS (BIND), DHCP, nftables, Snort IDS/IPS, OpenSSH, PAM, Active Directory

Embedded & Hardware RISC-V, ATmega2560, ATmega328P, Arduino, Raspberry Pi, PID control, serial protocols

Data & Practice PostgreSQL, Git, backup & disaster recovery, technical documentation, LLM/agentic AI tooling

Digital Art Blender 3D modeling, UV unwrapping, 8-bit RGB texture painting, Krita/Blender sync plugin development

PROJECTS

• GEO Calibration Marketing Campaign HUMAN-MADE

Campaign for NCSLI Symposium 2026 — the world's largest trade show in measurement & metrology.

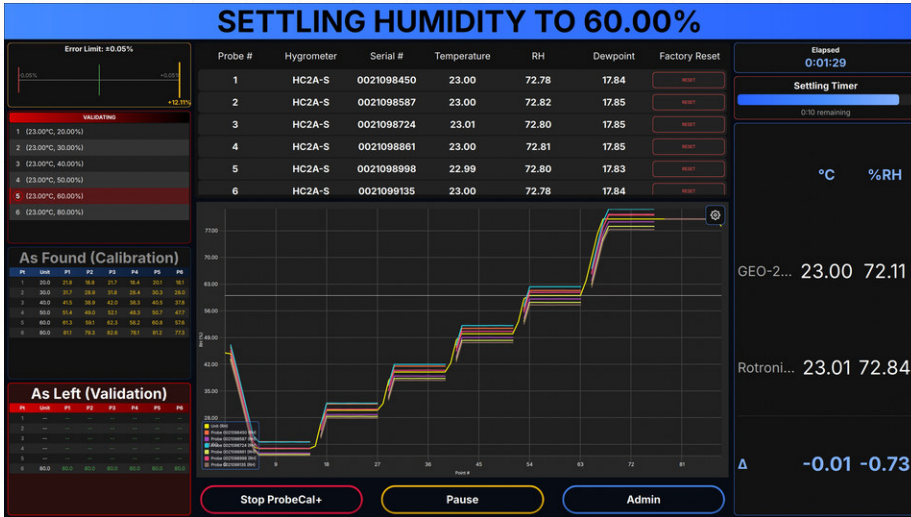
• GEO System AI-ASSISTED

Internal ERP for asset management, unit history, and calibration procedures. C++/Qt 6, PostgreSQL.

• RFA_Archiver AI-ASSISTED

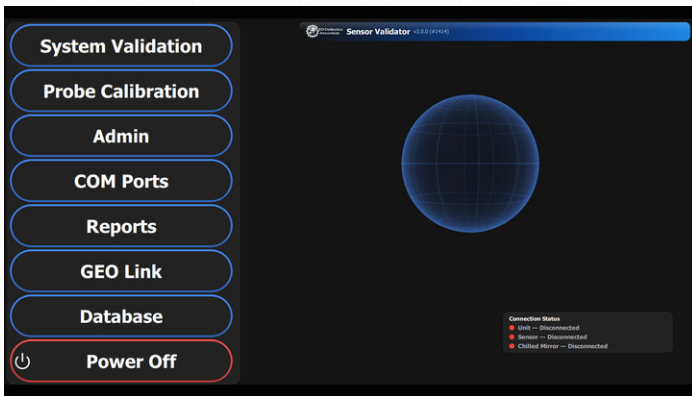
Customer database for mechanic shops: repair orders, invoicing, and record archiving.

SHOWCASE



GEO Validator X

Sole developer. C++/Qt 6 and QML — real-time multi-probe graphing, as-found and as-left calibration tables, and Excel reporting.



Sensor Validator

Sole developer. C++/Qt 6 — multi-channel environmental validation.



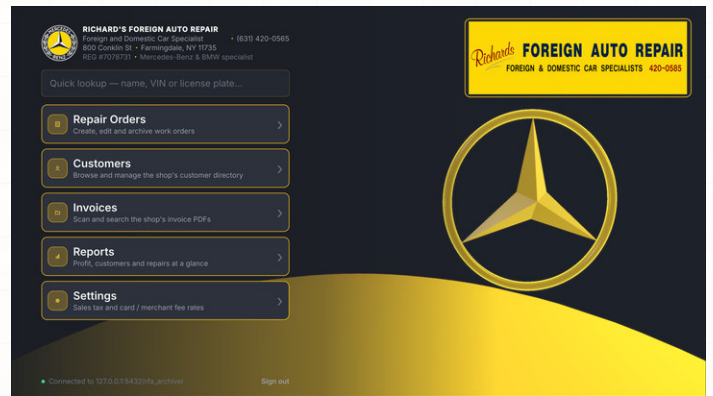
GEO Calibration Campaign

NCSLI Symposium 2026.



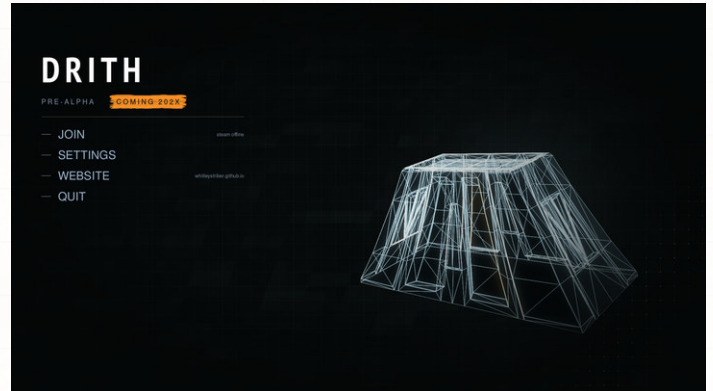
Drith -- Overworld Map

Environment and lighting in Unreal Engine 5.



RFA_Archiver

Repair orders, invoicing and record archiving for a mechanic shop.



Drith -- Launcher

Interface in Godot, over the game's own geometry.



Drith -- Gun Platform

Modelled and textured in Blender.